

Design of a 400 Gb/s Silicon Photonics Transceiver

For IEEE 802.3 DR4 Intra-DC Interconnects

Omar Mahfouz · Farah Elsayed · Mark Zakria · Mahmoud Magdy
Islam Ibrahim · Alyedeen Hassan · Khalid Khalil

Department of Electronics and Communications Engineering
Faculty of Engineering
Alexandria University

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Motivation and Targets

- Intra-datacenter PAM-4 receivers need **high responsivity**, **very low dark current**, and enough bandwidth for 53.125 GBd PAM-4 lanes.
- A **Ge-on-Si vertical n-i-p photodiode** offers a strong fit with standard SOI photonics.
- The chapter benchmarks the design against Yang Shi et al., *Photonics Research* 12(1), 2024.

Metric	Target
Responsivity	$\geq 0.95 \text{ A W}^{-1}$
Dark current	$\leq 1.3 \text{ nA}$
Bandwidth	$\geq 103 \text{ GHz}$
Detectivity	$\geq 2.95 \times 10^{10} \text{ Jones}$

- Vertical **n-i-p** stack on a 220 nm **SOI** platform.
- 50 nm **n++ Ge** top contact layer.
- 350 nm **intrinsic Ge** absorber.
- **p++ Si** platform for the anode path.
- **U-shaped electrode** reduces the RC penalty while preserving carrier collection.

Layer	Thickness
n++ Ge	50 nm
i-Ge	350 nm
p++ Si	220 nm
BOX	2 μ m

Optical Simulation Flow

- **Lumerical FDTD** models the taper-coupled optical absorption at 1310 nm.
- The simulation exports **optical generation rate** into CHARGE.
- Ge length is optimized to reach high absorption with minimum footprint.

Key optical design points:

- TE-driven O-band operation
- Adiabatic spot-size converter (500 nm $\rightarrow$ 5 μ m over 40 μ m)
- 3D generation-rate export: $G = P_{\text{abs}}/(h\nu)$

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Electrical Simulation Flow

- **Lumerical DEVICE CHARGE** solves carrier transport under reverse bias.
- SRH recombination enabled with the implemented Ge lifetime $\tau_n = \tau_p = 1$ ns.
- Fine mesh override: 20 nm max edge length over the Ge active region.
- Extracted quantities include:
 - ▶ Dark current and illuminated current
 - ▶ Band diagram, carrier density
 - ▶ Electric field and potential
- Bandwidth estimated from **transit-time + RC** limits.

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figures/phys_carrier_density.png

Dark Current Result

- Solver outputs drive the device metrics; literature values are used only as benchmarks.
- Low dark current is essential for:
 - ▶ Receiver sensitivity
 - ▶ Noise performance
 - ▶ Robust PAM-4 eye opening

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Internal Device Physics

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Band alignment, carrier density, and depletion shaping near $V = -1\text{ V}$ explain the tradeoff between responsivity, dark current, and speed.

Electric Field and Bandwidth

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- Transit-time and RC contributions are combined into the bandwidth model.
- All parasitic parameters (R_s , C_j , L_p , C_p) derived from geometry.

System-Level Receiver View

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figures/system_photocurrent_eye.png

- The repo extends the device into a **MATLAB PAM-4 receiver model** at 53.125 GBd.
- All device parameters flow from the CML .mat generated by device-level postprocessing.

PAM-4 Signal Quality

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- Matched-filter output shows **4 distinct PAM-4 levels**.
- Decision thresholds separate the symbol clusters.
- SNR, BER, and SER are computed from the simulation.
- The PD can now be studied both as a device and as a circuit block.

INTERCONNECT-Ready Compact Model

- The repo exports a **Lumerical INTERCONNECT-ready** photodetector data package.
- Generated artifacts include:
 - ▶ Responsivity table (frequency-dependent)
 - ▶ Bandwidth table (bias-dependent)
 - ▶ Dark-current table
 - ▶ Compact-model .mat source data
 - ▶ Equivalent-circuit parameters CSV

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Data Pipeline Overview

FDTD .lsf → fdtddata.mat → CHARGE .lsf → charge_data.mat → postprocess.m → CML .mat
→ system-level

All parameters flow from Lumerical solvers → MAT files → MATLAB postprocess → CML export → system-level scripts. **Nothing is hardcoded in the pipeline.**

Final Takeaways

- The chapter demonstrates a **Ge-on-Si O-band photodiode** targeting published high-speed metrics.
- The workflow is consistent across:
 - ▶ **Device-level FDTD**—optical absorption, generation rate
 - ▶ **Device-level CHARGE**—dark current, responsivity, band structure
 - ▶ **Device-level postprocess**—CML dataset, all derived parameters
 - ▶ **System-level MATLAB**—PAM-4 BER, eye diagrams, histograms
 - ▶ **INTERCONNECT-ready export**—circuit-level integration
- **Zero hardcoded paper values**—all metrics from solvers and geometry.